PULSE Instrument Cluster

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Document control

Field	Value
Document ID	PULSE-RTM-001
Title	Requirements Traceability Matrix
Generated	2026-09-08 by `scripts/gen_rtm.py`
Status	Derived; regenerate, never hand-edit
Sources	PULSE-StRS-001, PULSE-SyRS-001, PULSE-SAD-001, PULSE-SAF-001, PULSE-SEC-001

This matrix is produced from the ID tables in the source documents. A requirement's status, verification method and evidence are copied from its owning document. A story's trace list is the only link authored by hand. Dangling references fail the generator, so a matrix that exists is a matrix that is consistent.

1. Status summary

Group	Implemented	Partial	Planned S2	Planned S3	Planned S4	Total
Functional (FR)	15	5	0	1	1	22
Interface (IR)	4	1	0	0	0	5
Non-functional (NFR)	4	2	0	2	0	8
Functional safety (SR)	4	4	1	1	1	11
Cybersecurity (CS)	3	4	1	2	1	11
All	**30**	**16**	**2**	**6**	**3**	**57**

Coverage: 36 of 57 requirements are reached by at least one user story; 21 are traced only through their owning document (listed in section 5).

2. Stakeholder needs to stories to requirements

Need	Statement	Stories	Requirements reached
SN-1	Decide the cluster toolkit on measured evidence, with application cost separated from platform cost	US-5, US-8, US-10	FR-7, FR-12, FR-13, FR-15, FR-16, FR-17, NFR-3, NFR-7
SN-2	Build in-house competence in the hybrid cockpit pattern: one broker, a Linux cluster, an Android surface	US-2, US-8	FR-2, FR-3, FR-12, FR-13
SN-3	An HMI decoupled from the vehicle bus so a platform change does not restart HMI work	US-1, US-3	FR-1, FR-5, FR-8, FR-11
SN-4	A demonstrator presentable to a non-technical audience in under five minutes	US-4, US-6, US-7, US-9	FR-6, FR-9, FR-10, FR-14, IR-3, IR-4, NFR-2
SN-5	Competence that outlives the project: a new engineer brings the stack up from documentation alone	US-11	NFR-5
SN-6	Compliance readiness: the safety and security artefacts a production programme would ask for, produced while building rather than retrofitted	US-14, US-15, US-16, US-17	CS-1, CS-2, CS-3, CS-6, SR-1, SR-2, SR-3, SR-9, SR-11
SN-7	A closed loop, not a dashboard light: perception feeds a vehicle reaction through the same signal path the cluster renders	US-12, US-13	FR-18, FR-19, FR-20, FR-21, SR-7

3. Requirement to story, verification, status and evidence

3.1 Functional (FR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-1	The system shall represent all vehicle data using an open, standardised signal catalogue (COVESA VSS), not proprietary or bus-specific identifiers.	US-1	I	Implemented	`vss/vss_agl.json` compiled from VSS 6.0; 1,302 leaf signals
FR-2	The system shall extend the standard catalogue for domain-specific signals using the standard overlay mechanism only, with no forks of the upstream specification.	US-2	I	Implemented	`vss/agl_vss_overlay.vspec`, `vss/pulse_vss_overlay.vspec`, `vss/dms_vss_overlay.vspec`; upstream spec consumed unmodified
FR-3	The compiled signal tree shall be reproducible from versioned source files by a documented command.	US-2	T	Implemented	README “Regenerate the VSS JSON”; byte-identical regeneration checked by the regression suite (`vss-regeneration`), report in `docs/evidence/`
FR-4	A central broker shall serve the signal tree and allow multiple independent clients to subscribe concurrently.	none	D	Implemented	KUKSA databroker 0.7.0; Flutter cluster, Compose cluster and engine-audio service subscribe concurrently; suite check `broker-up`
FR-5	Signal producers and consumers shall be mutually decoupled: neither needs knowledge of the other’s implementation.	US-3	I	Implemented	Providers and HMIs share no code, only VSS paths (PULSE-SAD-001 section 4)

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-6	The system shall provide a telemetry source that generates physically plausible, correlated vehicle behaviour (speed, engine speed, gear, temperature, fuel, odometer) without any vehicle hardware.	US-4	D	Implemented	`emulator/telemetry_sim.py`: lap and cruise cycles with correlated RPM, fuel and odometer. Suite check `live-telemetry` passes
FR-7	The telemetry source shall run a repeatable cycle so that UI behaviour can be compared across runs and across implementations.	US-5	T	Partial	Phase table is fixed with no randomness, but sampling is wall-clock driven, so runs differ in timing. Fixed-step replay in S4
FR-8	The telemetry source shall be replaceable by a real vehicle-bus provider without modification to any HMI code.	US-3	T	Partial	Met by design (HMIs subscribe to VSS paths only); verified when the virtual CAN feeder replaces the simulator in S3
FR-9	The HMI shall display, updating live: road speed, engine speed with warning and redline indication, selected gear, fuel level and range, per-corner tyre pressures, and driver-relevant status…	US-6	D	Implemented	`pulse-cluster/lib/screen/pulse_screen.dart`; subscribed paths listed in PULSE-SAD-001 appendix A
FR-10	The HMI shall display motorsport telemetry: current lap number, current and best lap time, aerodynamic (DRS) state, and energy-recovery (ERS) level and state.	US-7	D	Implemented	`Vehicle.Motorsport.*` overlay signals rendered by both HMIs
FR-11	The HMI shall subscribe to abstract signals only. It shall not contain CAN frame parsing, DBC knowledge, or hardware-specific decoding.	US-1	T	Implemented	Regression suite check `hmi-purity` scans both HMI sources for CAN, DBC and SocketCAN tokens; passing, report in `docs/evidence/`
FR-12	The HMI shall be implemented twice, once in Flutter for Linux and once in Compose for Android Automotive, from a single shared visual design.	US-8	I	Implemented	`pulse-cluster/` (Flutter) and `pulse-cluster-android/` (Compose); design source in `design/`
FR-13	Both implementations shall consume the identical signal source, so any observed difference is attributable to the platform, not the data.	US-8	D	Partial	Same broker and protocol for both. Signal sets have diverged: Flutter subscribes to 28 paths, Compose to 17. The nine extra are the driver-monitoring and…
FR-14	The HMI shall render full-screen on the target strip display, and shall degrade gracefully to a normal window when that display is absent.	US-9	D	Implemented	`run.sh` locates a 2560 x 720 output at runtime and passes `CLUSTER_GEOM`; unset means a normal window
FR-15	The system shall support measurement of both implementations under identical telemetry: CPU utilisation, memory footprint, and shipped artefact size.	US-10	T	Partial	Measured once by hand: 123 vs 46 MB memory, 12.2 vs 11.5 % CPU, 50 vs 23 MB artefact (`BRIEF.md`). Scripted benchmark in S4

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-16	Measurements shall distinguish application-level cost from platform-level cost, since these drive different procurement decisions.	US-10	A	Partial	Platform memory reported separately (AAOS 3.2 to 4.0 GB from emulator); target-hardware figures planned S3 to S4
FR-17	Results shall be documented such that a reviewer can reproduce them from the repository.	US-10	T	Planned S4	Benchmark script and results file to be committed under `docs/`
FR-18	The system shall derive driver fatigue and distraction levels from a driver-facing camera using inference executed entirely on the local device, and publish them as VSS driver signals.	US-12	D	Implemented	`emulator/dms.py`: MediaPipe FaceLandmarker on CPU; publishes `Vehicle.Driver.FatigueLevel`, `DistractionLevel`, `AttentiveProbability`, `IsEyesOnRoad` at 10 Hz
FR-19	The system shall maintain a driver alert state of NONE, DROWSY, DISTRACTED or NO_FACE. It shall raise an alert only after a configurable hold time, and release it only after a configurable clear…	US-12	T	Implemented	`DMS_ALERT_HOLD_S` 3.0 s, `DMS_ALERT_CLEAR_S` 2.0 s; published as `Vehicle.Driver.Monitoring.AlertState`
FR-20	When cruise mode is active and the alert state is DROWSY, the vehicle model shall execute the minimum-risk manoeuvre specified in PULSE-SAF-001 SM-3, which the driver can abort by recovering…	US-13	T	Implemented	`telemetry_sim.py` cruise mode implements SM-3; published as `Vehicle.Driver.Monitoring.InterventionState` and `InterventionCountdown`
FR-21	The HMI shall render the driver attention level, the active alert as a full-width banner distinguishable by colour per alert type, and the intervention countdown.	US-12	D	Implemented	Attention panel and banner in `pulse_screen.dart`; attention meter in `classic_screen.dart`
FR-22	The system shall accept an autonomy-stack provider that publishes planned trajectory, detected objects and drive mode into the broker as VSS signals, so that the cluster can render an autonomy state…	none	D	Planned S3	Autoware planning-simulation bridge, work package 04; VSS overlay for autonomy signals to be added under `vss/`

3.2 Interface (IR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
IR-1	Client-to-broker communication shall use a language-neutral RPC interface supporting streaming subscriptions.	none	I	Implemented	gRPC over HTTP/2, `kuksa.val.v1` Subscribe stream; Dart and Kotlin stubs generated from the same `.proto`
IR-2	Signal semantics (units, ranges, allowed enumeration values, and encodings) shall be defined in the signal catalogue, not in client code.	none	I	Implemented	Units, `allowed` lists and `max` in the `.vspec` overlays; broker rejects out-of-catalogue values on set; suite check `unit-sanity` catches a value published…
IR-3	The target display interface shall be a 2560 x 720 automotive strip panel.	US-9	D	Implemented	Corsair Xeneon Edge on the desk rig; layout designed at 2560 x 720

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
IR-4	Display placement shall be configurable at runtime, not fixed at compilation.	US-9	D	Implemented	`CLUSTER_GEOM` environment variable read by the Linux runner; `run.sh` derives it from `xrandr`
IR-5	The system shall run without transport security in the local development configuration, and the security posture shall be an explicit, documented choice.	none	I	Partial	`--insecure` set in `scripts/start-databroker.sh`; documented as a deviation in PULSE-SEC-001 section 5. Start-up warning (CS-3) planned S2

3.3 Non-functional (NFR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
NFR-1	Responsiveness: displayed values shall track the telemetry source with no perceptible lag at a glance.	none	D	Implemented	Fast signals at 10 Hz, one broker tick from publish to render; observed at the review demo
NFR-2	Legibility: primary values (speed, gear, engine speed) shall be readable in a sub-second glance, consistent with driver-distraction practice.	US-6	D	Implemented	Design reviewed on the strip; numerals sized for the 720 px height
NFR-3	Footprint: the Linux implementation shall be deployable on constrained embedded hardware; total platform footprint is a first-class evaluation criterion.	US-10	T	Planned S3	Raspberry Pi 5 baseline, work package 03, week 9
NFR-4	Portability: migration to a different SoC or to a Yocto/AGL image shall not require HMI rework.	none	A	Planned S3	Argument to be recorded with the Pi 5 baseline; Flutter embedder is the only platform-specific layer
NFR-5	Reproducibility: a new engineer shall be able to bring up the full stack on a clean machine from documentation alone.	US-11	T	Implemented	Clean-machine bring-up performed 2026-08-18 and recorded in ONBOARDING.md, six blockers closed
NFR-6	Openness: the stack shall be built from open-source components, with no proprietary runtime licence required for evaluation.	none	I	Implemented	KUKSA (Apache-2.0), VSS (MPL-2.0), Flutter (BSD), MediaPipe (Apache-2.0), AOSP; NOTICE file for the face model
NFR-7	Determinism: repeated runs of the same cycle shall produce the same signal sequence, so comparisons are valid.	US-5	T	Partial	Same as FR-7: value sequence is deterministic in shape, sample timing is not. Fixed-step replay planned S4
NFR-8	Maintainability: upstream components shall be consumed at pinned versions, not floating heads.	none	I	Partial	Broker pinned to `0.7.0` by tag and digest; `kuksa-client==0.5.2` and Python deps in `requirements.txt`; VSS at v6.0; Dart deps in `pubspec.lock`. Remaining...

3.4 Functional safety (SR), owner PULSE-SAF-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
SR-1	An item definition shall be written for the cluster and driver-monitoring function, naming the system boundary, the assumed vehicle context, and the interfaces crossing that boundary (ISO 26262-3).	US-15	I	Partial	Section 2 of this document
SR-2	A preliminary hazard analysis and risk assessment (HARA) shall be recorded for the displayed functions, with severity, exposure and controllability rated per malfunction and a resulting ASIL proposal.	US-15	I	Partial	Section 4; eight hazards rated; review in S2
SR-3	Signals whose corruption or loss could mislead the driver (road speed, gear, warning indicators, drowsiness alert) shall be identified as safety-relevant and listed explicitly.	US-15	I	Implemented	Section 3
SR-4	The system shall detect loss of a safety-relevant signal, including broker disconnection and stale values, and shall indicate the degraded state rather than continuing to display the last known value.	none	T	Planned S2	SM-1; Flutter service reconnects but shows last value today
SR-5	Every safety-relevant signal shall carry a freshness criterion; a value older than its criterion shall be treated as invalid.	none	T	Partial	Criteria proposed in section 3; enforcement with SM-1 in S2
SR-6	Safety-relevant display elements shall fail visibly. A rendering fault shall not present a plausible but wrong value.	none	T	Planned S3	SM-2
SR-7	The minimum-risk manoeuvre triggered by sustained driver drowsiness shall be specified as a safety mechanism: trigger condition, driver-recovery window, escalation, and abort condition.	US-13	D	Implemented	SM-3; implemented in `emulator/telemetry_sim.py` cruise mode; demonstrated live
SR-8	Freedom from interference shall be argued for the mixed-criticality cockpit: infotainment content shall not be able to degrade or obscure the safety-relevant cluster surface.	none	A	Planned S4	Section 6 outline
SR-9	A safety-relevant requirement shall be traceable to the verification that demonstrates it, and each verification result shall be reproducible from the repository.	US-17	T	Partial	PULSE-RTM-001 generated from this table; regression suite runs before each demo and commits its report to `docs/evidence/`; safety-specific tests (SR-4 to…
SR-10	Any decision that would block a later ASIL argument, such as the choice of graphics stack, operating system or toolkit, shall be recorded as a documented deviation with its rationale.	none	I	Implemented	PULSE-SAD-001 section 8, DD-1 to DD-11

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
SR-11	A qualification gap list shall be maintained: for each component, what a production programme would still need (qualified toolchain, certified OS, evidence of a safety-qualified renderer).	US-15	I	Implemented	Section 8

3.5 Cybersecurity (CS), owner PULSE-SEC-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
CS-1	A threat analysis and risk assessment (TARA) shall be recorded for the signal path, covering the broker, both HMI clients, the telemetry provider and the camera input. Each threat scenario shall…	US-16	I	Partial	Section 4, nine scenarios
CS-2	The item boundary and its trust boundaries shall be documented, distinguishing on-device interfaces from anything reachable off the device.	US-16	I	Implemented	Section 2
CS-3	Broker communication shall support authenticated, encrypted transport (mutual TLS). Running without it shall be a configuration reserved for local development and flagged at start-up.	US-16	T	Planned S2	Section 5; `--insecure` in `scripts/start-databroker.sh` today
CS-4	Write access to the signal tree shall be authorised per client. A consumer shall not be able to publish, and no client shall write outside its declared signal set.	none	T	Planned S3	Section 5
CS-5	Signal values received from any provider shall be range- and type-validated against the catalogue before display, so a compromised provider cannot drive arbitrary content onto the cluster.	none	T	Partial	Broker-side validation implemented by KUKSA; HMI-side check and test in S2
CS-6	Camera frames shall not leave the device and shall not be persisted. Only derived state, such as the drowsiness level, shall be published.	US-14	I	Implemented	`emulator/dms.py` publishes six derived signals and writes no files; test in S2
CS-7	Security-relevant events, including authentication failure, unauthorised write attempt and client disconnection, shall be logged with a timestamp and without personal data.	none	T	Planned S3	Section 5
CS-8	All third-party dependencies shall be pinned and inventoried as a software bill of materials, and the inventory shall be checkable against published vulnerabilities.	none	T	Partial	Python, Dart and the broker image pinned. No SBOM yet. S2

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
CS-9	No credential, key or certificate shall be committed to the repository; the development configuration shall use clearly marked non-production material.	none	T	Implemented	Suite check `secrets` scans every tracked file for key and credential patterns; passing
CS-10	An update concept shall be described for the cluster and monitor software. It shall cover package authenticity and the behaviour on a failed update, as UNECE R155 and R156 require for type approval.	none	I	Planned S4	Section 6 outline
CS-11	Cybersecurity requirements shall be verified as part of the sprint regression suite, not as a separate end-of-project activity.	none	T	Partial	Suite exists (`scripts/regression.sh`) with range and secret checks for CS-5 and CS-9; CS-6 and CS-8 checks join in S2

4. Story to requirements

Story	Statement	Needs	Requirements	Verification of those requirements
US-1	As an HMI developer, I want every vehicle value available as a named VSS signal, so that I never touch bus formats or…	SN-3	FR-1, FR-11	I, T
US-2	As a platform engineer, I want domain signals added as VSS overlays, so that we stay compatible with the upstream…	SN-2	FR-2, FR-3	I, T
US-3	As an integrator, I want producers and consumers to meet only at the broker, so that either side can be swapped…	SN-3	FR-5, FR-8	I, T
US-4	As a demonstrator, I want a realistic drive cycle without a vehicle, so that the cluster behaves plausibly on a desk.	SN-4	FR-6	D
US-5	As an evaluator, I want the same cycle to replay identically, so that measurements of the two HMIs are comparable.	SN-1	FR-7, NFR-7	T
US-6	As a driver, I want speed, RPM, gear, fuel and tyre pressures at a glance, so that I can read the vehicle state in…	SN-4	FR-9, NFR-2	D
US-7	As a race engineer, I want lap, DRS and ERS state on the cluster, so that motorsport telemetry is visible live.	SN-4	FR-10	D
US-8	As a stakeholder, I want the same design built in Flutter and Compose from one signal source, so that the platform…	SN-1, SN-2	FR-12, FR-13	D, I
US-9	As a presenter, I want the cluster full-screen on the strip display but windowed elsewhere, so that the demo works on…	SN-4	FR-14, IR-3, IR-4	D
US-10	As the sponsor, I want application cost separated from platform cost, so that the toolkit decision and the hardware…	SN-1	FR-15, FR-16, FR-17, NFR-3	A, T
US-11	As a new engineer, I want to bring the stack up from documentation alone, so that the competence outlives the project.	SN-5	NFR-5	T

Story	Statement	Needs	Requirements	Verification of those requirements
US-12	As a driver, I want the cluster to warn me when I am drowsy or distracted, so that I regain attention before anything…	SN-7	FR-18, FR-19, FR-21	D, T
US-13	As a safety engineer, I want sustained drowsiness to trigger a defined minimum-risk manoeuvre with a recovery window…	SN-7	FR-20, SR-7	D, T
US-14	As a privacy officer, I want camera frames to stay on the device, so that driver monitoring never becomes surveillance.	SN-6	CS-6	I
US-15	As a safety manager, I want an item definition, hazard analysis and safety-relevant signal list for the cluster, so…	SN-6	SR-1, SR-2, SR-3, SR-11	I
US-16	As a security manager, I want a threat analysis and trust-boundary description for the signal path, so that the…	SN-6	CS-1, CS-2, CS-3	I, T
US-17	As a reviewer, I want every requirement traced to a story and a verification, so that I can see what the prototype…	SN-6	SR-9	T

5. Requirements not reached by a user story

These are traced only through their owning document and its verification column. Add a story or accept the gap at review.

ID	Requirement	Owner	Status (W3)
CS-4	Write access to the signal tree shall be authorised per client. A consumer shall not be able to publish, and no client shall write outside its declared signal set.	PULSE-SEC-001	Planned S3
CS-5	Signal values received from any provider shall be range- and type-validated against the catalogue before display, so a compromised provider cannot drive arbitrary content onto the cluster.	PULSE-SEC-001	Partial
CS-7	Security-relevant events, including authentication failure, unauthorised write attempt and client disconnection, shall be logged with a timestamp and without personal data.	PULSE-SEC-001	Planned S3
CS-8	All third-party dependencies shall be pinned and inventoried as a software bill of materials, and the inventory shall be checkable against published vulnerabilities.	PULSE-SEC-001	Partial
CS-9	No credential, key or certificate shall be committed to the repository; the development configuration shall use clearly marked non-production material.	PULSE-SEC-001	Implemented
CS-10	An update concept shall be described for the cluster and monitor software. It shall cover package authenticity and the behaviour on a failed update, as UNECE R155 and R156 require for type approval.	PULSE-SEC-001	Planned S4
CS-11	Cybersecurity requirements shall be verified as part of the sprint regression suite, not as a separate end-of-project activity.	PULSE-SEC-001	Partial
FR-4	A central broker shall serve the signal tree and allow multiple independent clients to subscribe concurrently.	PULSE-SyRS-001	Implemented
FR-22	The system shall accept an autonomy-stack provider that publishes planned trajectory, detected objects and drive mode into the broker as VSS signals, so that the cluster can render an autonomy state…	PULSE-SyRS-001	Planned S3
IR-1	Client-to-broker communication shall use a language-neutral RPC interface supporting streaming subscriptions.	PULSE-SyRS-001	Implemented
IR-2	Signal semantics (units, ranges, allowed enumeration values, and encodings) shall be defined in the signal catalogue, not in client code.	PULSE-SyRS-001	Implemented

ID	Requirement	Owner	Status (W3)
IR-5	The system shall run without transport security in the local development configuration, and the security posture shall be an explicit, documented choice.	PULSE-SyRS-001	Partial
NFR-1	Responsiveness: displayed values shall track the telemetry source with no perceptible lag at a glance.	PULSE-SyRS-001	Implemented
NFR-4	Portability: migration to a different SoC or to a Yocto/AGL image shall not require HMI rework.	PULSE-SyRS-001	Planned S3
NFR-6	Openness: the stack shall be built from open-source components, with no proprietary runtime licence required for evaluation.	PULSE-SyRS-001	Implemented
NFR-8	Maintainability: upstream components shall be consumed at pinned versions, not floating heads.	PULSE-SyRS-001	Partial
SR-4	The system shall detect loss of a safety-relevant signal, including broker disconnection and stale values, and shall indicate the degraded state rather than continuing to display the last known value.	PULSE-SAF-001	Planned S2
SR-5	Every safety-relevant signal shall carry a freshness criterion; a value older than its criterion shall be treated as invalid.	PULSE-SAF-001	Partial
SR-6	Safety-relevant display elements shall fail visibly. A rendering fault shall not present a plausible but wrong value.	PULSE-SAF-001	Planned S3
SR-8	Freedom from interference shall be argued for the mixed-criticality cockpit: infotainment content shall not be able to degrade or obscure the safety-relevant cluster surface.	PULSE-SAF-001	Planned S4
SR-10	Any decision that would block a later ASIL argument, such as the choice of graphics stack, operating system or toolkit, shall be recorded as a documented deviation with its rationale.	PULSE-SAF-001	Implemented

6. Consistency

Needs: 7. Stories: 17. Requirements: 57. Dangling references: 0.